

A hybrid model for exchange rate prediction

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Abstract

Exchange rate forecasting is an important problem. Several forecasting techniques have been proposed in order to gain some advantages. Most of them are either as good as random walk forecasting models or slightly worse. Some researchers argued that this shows the efficiency of the exchange market. We propose a two stage forecasting model which incorporates parametric techniques such as autoregressive integrated moving average (ARIMA), vector autoregressive (VAR) and co-integration techniques, and nonparametric techniques such as support vector regression (SVR) and artificial neural networks (ANN). Comparison of these models showed that input selection is very important. Furthermore, our findings show that the SVR technique outperforms the ANN for two input selection methods.

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1. Introduction

Exchange rate forecasting is an important problem that has been studied by researchers and practitioners extensively. It is argued that exchange rate market is very efficient. Therefore, it is difficult to make short term and long term forecasting efficiently [2,13,25,38]. Several techniques have been proposed and applied to exchange rate forecasting and estimation of the volatility in order to beat the random walk model. These techniques can be put into the following categories.

The first group of techniques uses the economic theory to understand the structural relations between exchange rate and other variables and also statistical methods which try to identify the structure of the serial

correlation and nonlinearity in time series. They are in the family of parametric models. Econometric and time series models are widely applied to foreign exchange market. Several researchers criticized the forecasting performance of these techniques and some of them have found that the random walk model outperforms the econometric and time series techniques [18,25,38]. The reason is that most of the econometric models are linear and used under specific assumptions. For example, ARIMA models assume a linear relationship between the current value of the underlying variables and previous values of the variable and error terms. Time series models are highly nonlinear and the mean and variance of the series can change overtime. In order to overcome this difficulty, an autoregressive conditional heteroscedasticity (ARCH) model is introduced by Engle [16] and this model is generalized by Bollerslev [1]. Different implementation of GARCH models has been proposed in order to overcome some difficulties such as nonlinearities and long term memory [15,14].

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Fernandes tested whether the conditional heteroscedasticity models (CHM) capture the nonlinearities in the data. Nine out of twelve currencies, CHM models are good approximations. This suggests that CHM captures substantially but not completely the nonlinearities in the data [18]. Furthermore, instead of using GARCH models, their extension such as FIGARCH and IGARCH models are used for forecasting the market exchange rate. In a recent paper, Vilasuso [43] showed that the FIGARCH model is better than IGARCH and GARCH models for capturing the salient feature of the exchange rate volatility. The FIGARCH model generates superior out of sample volatility forecasting. Some researchers argued that CHM type models are better than random walk model and others showed that the random walk model is as good as CHMs or better than CHM [2,13]. The smooth transition autoregressive model (STAR) and exponential smooth transition autoregressive model (ESTAR) have been used to discover the dynamics of exchange rate forecasting. The ESTAR model shows strong predictability at horizons of 2 to 3 years. However, it does not have this predictability for shorter horizons [25].

Nonparametric models have been used extensively the last decade. The reason is the development of new techniques in artificial intelligence and increasing power of computers. These techniques have been applied to several areas such as stock price prediction, option pricing, and credit risk scoring [5–7,19,22,23,26,30,36,37,39]. Most widely used techniques are the multilayer perceptron (MLP), radial basis function (RBF) networks and recurrent networks. Because of the efficiency of the foreign exchange market it is difficult to use statistical methods to forecast the dynamic behavior of the time series. Therefore, it is not wise to use linear models. RBF networks were applied to \$US/\$NZ exchange rate forecasting and showed that RBF network outperforms the linear autoregressive (LAR) models [44]. In their paper, Yao and Tan compared the neural network model with technical forecasting for Swiss Franc and American Dollar and they concluded that it is not easy to forecast if the market is efficient [45]. In addition to this, ANN and chaotic models were compared with random walk model. Lisi and Schiavo [29] indicated that ANN and chaotic models outperform the random walk model. In the exchange rate market forecasting literature, neural networks have been used and argued by several researchers (for examples see Refs. [6,7,11,12,26,29,30,44,45]).

Recently, in classification and regression, a novel method which is called support vector machine (SVM)

has been developed by Vapnik and is used successfully in several classification and regression problems [9,42]. SVM uses the structural risk minimization theory [3,9,34] unlike other methods such as ANN and ARIMA that employ the empirical risk minimization theory. It has been shown that the SVM problem is a convex optimization problem which means that the optimal solution is global. On the other hand, an ANN model uses the backpropagation or its variant algorithm to find the optimal weights. The problem is nonconvex and its solution is in one of the local minima. Because of this, theoretically, the SVM algorithm is superior to the ANN model. This has also been proven experimentally [33,35]. Support vector regression (SVR) has been used to predict stock market indices such as NASDAQ and Dow Jones, and short term stock prices [23,36,37].

Our objective is to develop a two stage forecasting model. In the first stage, we propose an input selection process by using time series models such as autoregressive integrated moving average (ARIMA) and co-integration analysis. After determining the number of inputs in the first stage, we apply state of the art techniques, namely ANN and SVR. In this way, powerful side of time series models and artificial intelligence model are discovered. Since the ANN and SVR technique are data-driven, it is very crucial to determine the right inputs. In the estimation process, ANN, and SVR do not use any assumption regarding the distribution of the data. On the other hand, time series techniques use several assumptions about the data. Therefore, estimation and forecasting are based on these assumptions. This is a very important drawback. In order to overcome these difficulties, we propose a two stage algorithm.

The remainder of this paper is organized as follows. In Section 2, we explain the methodology. Section 3 gives experimental results. Finally, Section 4 concludes the paper.

2. Methodology

In this section, co-integration method, SVR and ANN models will be explained briefly. The first method is used to identify the relationship between dependent and independent variables which can be the lagged values of dependent variables or exogenous variables. Then, two machine learning techniques, multilayer perceptron (MLP), which uses the back-propagation algorithm [21] as a training algorithm and SVR will be applied to the model that we specified in the first stage.

2.1. Time series analysis

Two techniques can be adopted to determine the inputs of a forecasting model. One of them is to use the autoregressive integrated moving average (ARIMA) model which tries to find the optimal number of previous values of the dependent variable and random shocks. The second one is a vector autoregressive (VAR) model and co-integration analysis. Next we briefly explain these two methods.

The general autoregressive moving average (ARMA) model of a time series Y can be written as

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} \quad (1)$$

where ε_t is independent and identically distributed with mean 0 and variance σ^2 . This means there is a linear relationship among the past values and future values of Y and random shocks. In order to determine the order of ARIMA model, ACF and partial ACF are used in conjunction with the Akaike information (AIC) or Schwarz Bayes information (BIC) criterion. Augmented Dickey Fuller (ADF) test can be conducted to test the stationarity. The ARIMA model follows an iterative process of model identification, parameter estimation and diagnostic checking. More information in ARIMA models can be found in Ref. [20].

Co-integration analysis, introduced by Granger, is the determination of long run relationships in economics. The basic idea behind the co-integration is that, if two or more series move closely together, even though the series themselves are trended, the difference between them is constant [15].

Consider the following vector error correction model of order p :

$$\Delta y_t = \alpha \beta' y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + \varepsilon_t \quad (2)$$

where y_t is an $(n \times 1)$ vector of $I(1)$ variables, $\alpha \beta'$ is an $(n \times n)$ matrix such that the $(n \times r)$ matrices have rank r , $\Gamma_j, j = 1, 2, \dots, p-1$, are $(n \times n)$ parameter matrices, and ε_t is an $(n \times 1)$ vector of white noise with a positive definite covariance matrix. If $0 < r < n$, the variables in y_t are co-integrated with r co-integrating relationships βy_t .

In order to test the null hypothesis of no co-integration between the set of $I(1)$ variables, the ordinary least squares (OLS) method can be used to estimate the parameters and then, the unit root test can be applied to residuals. Rejecting the null hypothesis of a unit root

test is evidence in favor of co-integration. Johansen [24] proposed the maximum likelihood method to estimate the β which can be derived as the solution of a generalized eigenvalue problem. Likelihood ratio tests of hypotheses about the number of co-integrating vectors can be based on these eigenvalues.

Our goal is to use the co-integration analysis to determine the order of the integration between variables if the co-integration exists. The major advantage of the time series models would be the identification of the independent variables in our models which will be explained next. We will employ the parametric models to choose the influential variables or previous values of the dependent variables.

2.2. Support vector regression

SVMs for classification and regression based on structural risk minimization are developed by Vapnik [42]. In the ε -insensitive SVR, our goal is to find a function $f(x)$ that has an ε deviation from the actually obtained target y_i for all training data and at the same time is as flat as possible. Suppose $f(x)$ takes the following form:

$$f(x) = wx + b \quad \mathbf{w} \in X, b \in \mathfrak{R}. \quad (3)$$

In the case where the constraints are infeasible, we introduce slack variables ξ_i, ξ_i^* . This case is called the soft margin formulation, and is described by the following problem.

$$\begin{aligned} \min & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \\ \text{Subject to} & \\ & y_i - \mathbf{w} \mathbf{x}_i - b \leq \varepsilon + \xi_i \\ & \mathbf{w} \mathbf{x}_i + b - y_i \leq \varepsilon + \xi_i^* \\ & \xi_i, \xi_i^* \geq 0 \\ & C > 0 \end{aligned} \quad (4)$$

where, C determines the trade-off between the flatness of the $f(x)$ and the amount up to which deviations larger than ε are tolerated. Note that, ξ_i, ξ_i^* are called slack variables.

In order to solve problem (4), we formulate the dual problem by constructing the Lagrange function. The dual problem of Eq. (4) becomes:

$$\begin{aligned} \max & -\frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l (\lambda_i - \lambda_i^*) (\lambda_j - \lambda_j^*) \mathbf{x}_i \mathbf{x}_j - \varepsilon \sum_{i=1}^l (\lambda_i + \lambda_i^*) + \sum_{i=1}^l y_i (\lambda_i - \lambda_i^*) \\ \text{Subject to} & \sum (\lambda_i - \lambda_i^*) = 0 \\ & \lambda_i, \lambda_i^* \in (0, C). \end{aligned} \quad (5)$$

Solving for $\mathbf{w}$, we have

$$\mathbf{w}^* = \sum_{i=1}^l (\lambda_i - \lambda_i^*) \mathbf{x}_i, \quad (6)$$

$$f(x) = \sum_{i=1}^l (\lambda_i - \lambda_i^*) \mathbf{x}_i \mathbf{x} + b^*.$$

So far, we have explained the linear SVR. Let us look at the nonlinear case briefly. First of all, we need to map the input space into the feature space and try to find a linear regression hyperplane in the feature space. Using the trick of kernel functions [3,9,33,34], we have the following QP problem

$$\begin{aligned} \max & -\frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l (\lambda_i - \lambda_i^*) (\lambda_j - \lambda_j^*) K(\mathbf{x}_i, \mathbf{x}_j) - \varepsilon \sum_{i=1}^l (\lambda_i + \lambda_i^*) + \sum_{i=1}^l \nu_i (\lambda_i - \lambda_i^*) \\ \text{Subject to } & \sum (\lambda_i - \lambda_i^*) = 0 \\ & \lambda_i, \lambda_i^* \in (0, C) \end{aligned} \quad (7)$$

At the optimal solution, we obtain

$$\begin{aligned} \mathbf{w}^* &= \sum_{i=1}^l (\lambda_i - \lambda_i^*) K(\mathbf{x}, \mathbf{x}_i), \text{ and} \\ f(x) &= \sum_{i=1}^l (\lambda_i - \lambda_i^*) K(\mathbf{x}, \mathbf{x}_i) + b, \end{aligned} \quad (8)$$

where $K(.,.)$ is a kernel function.

According to Refs. [3,9,42], any symmetric positive semi-definite function, which satisfies Mercer's conditions, can be used as a kernel function in the SVMs context. Usually we have more than one kernel to map the input space into the feature space. The question is which kernel functions provide good generalization for a particular problem. We could not say that one kernel outperforms the others. Therefore, one has to use more than one kernel function for a particular problem. Some validation techniques such as bootstrapping and cross-validation can be used to determine a good kernel [4,10]. Even when we decide for a kernel function, we have to decide what are the parameters of the kernel. For instance, RBF kernel has a parameter σ and one has to decide the value of σ before the experiment. Selection of this parameter is very important. Many algorithms are proposed to solve the SVM optimization problem [8,10,27,31,32,34]. We can divide these algorithms into two groups: (1) classical nonlinear algorithms such as gradient descent/ascent algorithms, and Zoutendijk's methods [3,34]; and (2) state of the art interior point algorithms such as the primal dual path following algorithm [40,41]. One can solve moderate sizes of problems with these algorithms. In order to achieve the expected efficiency with a large-scale

problem, decomposition techniques must be applied [8,31,32].

2.3. Artificial neural networks

Neural networks have recently gained popularity to explore the dynamics of a variety of financial applications [11,19,22,26,28,30,44,45]. Since the exchange markets are highly volatile, complex with noise market conditions, several neural network models such as multilayer perceptron (MLP), radial basis function networks (RBF) and recurrent networks have been applied to exchange rate forecasting [11,30,45].

MLP networks have the capability of a complex mapping between input and output that enables the network to approximate nonlinear functions. Consider a two-layer MLP network consisting of n input, and a hidden layer of s hidden neurons and a layer of m output neurons. MLP networks operate as follows: input units receive a pattern vector, $\mathbf{x} = (x_1, x_2, \dots, x_n)$ from an external world, which is propagated to all units in the hidden layer. Every hidden neuron j first computes the net input $h_j = \sum_i w_{ij} x_i$ and then produces as output $V_j = f(h_j) = f(\sum_i w_{ij} x_i)$ where f is a differentiable transfer function [21]. Each output unit k receives as input the output of the hidden layer and, repeating all the operations just described, we have

$$O_k = f\left(\sum_{j=1}^s V_j w_{jk}\right) = f\left(\sum_{j=1}^s f\left(\sum_{i=1}^n x_i w_{ij}\right) w_{jk}\right). \quad (9)$$

Several optimization algorithms can be used to train the MLP network. One of the most widely used training algorithms is the backpropagation algorithm (BP). The BP algorithm minimizes the total square error by using the general delta rule [21].

The challenging task is to determine the number of hidden layers, number of neurons in each layer, learning and momentum parameter. These parameters can be determined with trial and error or genetic algorithms can be used to find the optimal architecture of the network. Extensive information can be found in Refs. [30,35].

MLP networks can be used stand alone for exchange forecasting. The difficulty is to determine the right input and input size. This can be achieved by using the underlying economics theory to identify the influential variables. In addition to this, statistical techniques such as autoregressive models or co-inte-

Table 1
ARIMA models for each exchange rate

Exchange rates	ARIMA models
Euro/Dollar	ARIMA(3,1,0)
Pound/Dollar	ARIMA(4,1,0)
JPY/Dollar	ARIMA(3,1,0)
AUD/Dollar	ARIMA(2,1,0)

gration can help us to determine the input variables for MLP networks.

3. Experiments

Exchange rate forecasting is a difficult task due to changing dynamics of its driving factors. There is a large number of factors that influence the daily value of the exchange rates. They can be identified by using time series models such as ARIMA and VAR techniques.

Daily values of exchange rates for Euro/Dollar, Pound/Dollar, JPY/Dollar and AUD/Dollar were used—from January 1, 2000 to May 26, 2004. The data set was randomly divided into three groups, training, cross-validation, and testing set. The number of examples in each set is 1544, 100, and 60, respectively. Our analysis consists of two parts. First, we conduct a time series analysis to determine the number of inputs by using ARIMA method, VAR and co-integration analysis for Euro/Dollar and GBP/Dollar rates.

Two time series models have been used to identify the inputs. These techniques can be put in two categories, univariate time series, which are ARIMA models, and multivariate time series, VAR and co-integration techniques. After determining the number of inputs with these methods, SVR and MLP networks are applied. The performances of SVR and MLP networks are compared with each other in terms of mean square error (MSE) and mean absolute error (MAE). Furthermore, we compare the performance of the input selection method, namely ARIMA vs. VAR and co-integration techniques. The testing set, which is not seen by the learning algorithms before, is used for comparison. In the next two subsections, performance of learning algorithms (SVR and MLP networks) is given for input selection techniques.

Table 2
Mean square error and mean absolute error of MLP network for testing set

	EURO	GBP	JPY	AUD
MSE	0.0000294	0.0000115	0.0535000	0.0000017
MAE	0.0033000	0.0025000	0.1300000	0.0009735

Table 3
Mean square error and mean absolute error of SVR for testing set

	EURO	GBP	JPY	AUD
MSE	0.0000037	0.0000021	0.0106000	0.0000009
MAE	0.001400	0.001100	0.070900	0.000654

3.1. ARIMA method for input selection

The ADF unit root test revealed that Euro/Dollar, Pound/Dollar, JPY/Dollar and AUD/Dollar time series are not stationary. Based on autocorrelation and partial autocorrelation, the following models are selected for each exchange rate datasets (see Table 1).

Table 1 shows that the following functions give us the relationship between input and output for each time series.

$$\begin{aligned}x_t &= f(x_{t-1}, x_{t-2}, x_{t-3}), \\y_t &= f(y_{t-1}, y_{t-2}, y_{t-3}, y_{t-4}), \\z_t &= f(z_{t-1}, z_{t-2}, z_{t-3}), \\v_t &= f(v_{t-1}, v_{t-2}).\end{aligned}\quad (10)$$

where x = Euro/Dollar, y = Pound/Dollar, z = JPY/Dollar and v = AUD/Dollar.

MLP networks with the backpropagation algorithm have been applied for each series. Several studies showed that one hidden layer provides good generalization capability for financial forecasting problems. The number of hidden units influences the performance of the network. It is important not to over-fit the MLP networks with a large number of hidden units than can memorize the data. In order to avoid this problem, a 10-fold cross-validation technique is used to select the MLP architecture in terms of the validation error. MSE and MAE for the testing set are shown in Table 2 for Euro/Dollar, GBP/Dollar, JPY/Dollar and AUD/Dollar exchange rates.

Furthermore, the SVR technique was also used as an alternative forecasting method. Since SVR has some free parameters such as the kernel function and its parameters, we need to specify them before running the algorithm. Free parameters of the SVR method have been determined by using a 10-fold cross-validation technique. Optimal parameters are shown in Table 3

Table 4
Comparison of MLP and SVR method performance with t -test for testing set

Exchange rates	t -statistics	t -crit. value (0.05)	p -value
EURO/Dollar	3.8185	1.6626	0.0001
GBP/Dollar	4.5297	1.6626	0.0000
JPY/Dollar	2.5530	1.6626	0.0062
AUD/Dollar	2.6761	1.6626	0.0044

Table 5

Trace and maximum eigenvalue test statistics for various values of the co-integration rank r with critical values for a 1%

	Variables	Trace	Critical value	$\lambda_{\max}$	Critical value
$r \leq 0$	GBP/Dollar	60.699	62.520	28.653	36.193
$r \leq 1$	EURO/Dollar	58.921	41.081	35.845	29.263
$r \leq 2$	JPY/Dollar	38.076	16.162	26.432	21.747
$r \leq 3$	AUD/Dollar	7.571	6.635	7.571	6.635

with MSE and MAE of the testing set for each exchange rate.

We have conducted a t -test to see if there is a significant difference between MLP and SVR methods in the testing period. Null hypothesis is stated that there is no difference between the performance of MLP network and SVR method in terms of MSE when ARIMA input selection technique is used. For each series, we have concluded that the SVR method outperforms the MLP network based on t -test results (see Table 4).

3.2. Vector autoregression and co-integration method for input selection

We found that time series are $I(1)$ by using Augmented Dickey Fuller (ADF) test. We select the time lag length using the likelihood ratio test and Akaike information criteria (AIC). These tests yield the optimal time lag length of $k=4$ for each exchange rate. A model with a linear trend was assumed. The results of trace and maximum eigenvalue test statistics are given in Table 5. According to Table 5, there is no co-integration between the exchange rates. Because of this, we could not use co-integration analysis for the input selection process.

Because of the co-integration analysis results, we turn our attention to vector autoregressive models in order to determine the inputs. We chose the time lag length as $k=4$ using the likelihood ratio test and AIC.

Table 7

Mean square error (MSE) and mean absolute error (MAE) of MLP network for testing set by using VAR method for input selection

	EURO	GBP	JPY	AUD
MSE	0.0000049	0.0000091	0.0323000	0.0000056
MAE	0.0018000	0.0025000	0.1437000	0.0020000

Then a VAR model was assumed and the parameters of the model were estimated. Investigation of residuals by the Granger causality test [17] shows that the Euro/Dollar rate has a significant Granger-Causal impact on the GBP/Dollar rate and vice versa (see Table 6).

From these results, the relationship between the exchange rates can be explained by the following equations.

$$x_t = f(x_{t-1}, \dots, x_{t-4}, y_{t-1}, \dots, y_{t-4}) \quad (11a)$$

$$y_t = f(y_{t-1}, \dots, y_{t-4}, x_{t-1}, \dots, x_{t-4}) \quad (11b)$$

$$z_t = f(z_{t-1}, \dots, z_{t-4}, x_{t-1}, \dots, x_{t-4}, y_{t-1}, \dots, y_{t-4}) \quad (11c)$$

$$v_t = f(v_{t-1}, \dots, v_{t-4}, y_{t-1}, \dots, y_{t-4}) \quad (11d)$$

$$x_t : (\text{Euro/Dollar})_t$$

$$y_t : (\text{GBP/Dollar})_t$$

$$z_t : (\text{JPY/Dollar})_t$$

$$v_t : (\text{AUD/Dollar})_t.$$

From Table 6 and Eqs. (11a) (11b) (11c) (11d), we draw the following results: EURO/Dollar exchange rate depends on its previous values and GBP/Dollar rate's on four previous values. GBP/Dollar exchange rate can be determined by its previous values and EURO/Dollar

Table 6

Granger causality tests

	F -value	F -probability		F -value	F -probability
Equation 1: (11a)			Equation 3: (11c)		
GBP	2.4601	0.0420	GBP	3.4281	0.0085
EUR	10,482.2311	0.0000	EUR	0.6336	0.6386
USD/JPY	1.2071	0.3059	USD/JPY	27,977.3069	0.0000
USD/AUD	0.7783	0.5392	USD/AUD	3.063	0.0158
Equation 2: (11b)			Equation 4: (11d)		
GBP	7622.7929	0.0000	GBP	1.4386	0.2188
EUR	2.5388	0.0383	EUR	2.3721	0.0499
USD/JPY	1.1301	0.3406	USD/JPY	0.5233	0.7186
USD/AUD	0.7348	0.5682	USD/AUD	11,456.0016	0.0000

Table 8

Mean square error (MSE) and mean absolute error (MAE) of SVR for testing set by using VAR method for input selection

	EURO	GBP	JPY	AUD
MSE	0.0000028	0.0000122	0.0159000	0.0000040
MAE	0.001400	0.003200	0.098700	0.001600

rate's by its four previous values. JPY/Dollar rates can be determined by its four previous values, similarly EURO/Dollar and GBP/Dollar rates by its four previous values. We need four previous values of AUD/Dollar and GBP/Dollar to determine the AUD/Dollar exchange rate.

Next, we will use this information on SVR and MLP methods. We have used the same experimental design as we used for univariate analysis. An MLP network with one hidden layer is used for each series. The number of hidden units and other parameters are determined by using a 10-fold cross-validation. Performance of the MLP networks is shown in Table 7.

As we stated in Section 3.1, selection of the free parameters for the SVR method is very important. SVR is very sensitive to these parameters. For this reason, a 10-fold cross-validation technique is employed in order to determine the right parameters. After determining these, an optimal solution is found by solving the problem. The performance of the SVR method with the VAR input selection process is given in Table 8 for each exchange rate in terms of MSE and MAE.

If we compare the MSEs of MLP and SVR method, we see that SVR outperforms the MLP networks for VAR input selection method for all series except the GBP/Dollar exchange rate. We have also performed a *t*-test to check if SVR outperforms the MLP methods for Euro/Dollar, GBP/Dollar, JPY/Dollar and AUD/Dollar exchange rates. We test the null hypothesis that the performances of SVR and MLP methods are the same for each exchange rate. Test results show that we accept the alternative hypothesis which is, SVR method outperforms the MLP methods for Euro/Dollar, JPY/Dollar and AUD/Dollar except GBP/Dollar (see Table 9). The performance of SVR and MLP network is the same for GBP/Dollar. All individual *t*-tests are

Table 9

Comparison of MLP and SVR method performance with *t*-test for testing set

Exchange rates	<i>t</i> -statistics	<i>t</i> -critical value (0.05)	<i>p</i> -value
EURO/Dollar	2.0085	1.6672	0.0243
GBP/Dollar	−1.0043	1.6672	0.1594
JPY/Dollar	2.3954	1.6672	0.0097
AUD/Dollar	2.2953	1.6672	0.0124

Table 10

MSE of ARIMA and VAR input selection procedure for testing set

Exchange rates	MLP network		SVR method	
	ARIMA	VAR	ARIMA	VAR
EURO/Dollar	0.000029	0.000005	0.000004	0.000003
GBP/Dollar	0.000012	0.000009	0.000002	0.000012
JPY/Dollar	0.053500	0.032300	0.010600	0.015900
AUD/Dollar	0.000002	0.000006	0.000001	0.000004

performed at 5% significance level. This means that the SVR method outperforms the MLP networks for exchange rate forecasting.

3.3. Comparison of input selection techniques

Until this point, we have compared the SVR and MLP network. Comparison of the input selection process, namely ARIMA vs. VAR, would reveal some information. The following design is used for the comparison of input selection. Since we employ two methods after the input selection procedure, MSE and MAE of input selection process are given in the following table.

Table 10 shows the MSE of ARIMA and VAR technique for MLP network and SVR method. It is wise to use VAR technique in order to determine the input of the MLP networks for three exchange rates; EURO/Dollar, GBP/Dollar and JPY/Dollar. On the other hand, ARIMA outperforms the VAR technique to determine the input of the SVR method for the three exchange rates, GBP/Dollar, JPY/Dollar, and AUD/Dollar. As we know from Sections 3.1 and 3.2, SVR outperforms the MLP networks. Therefore, ARIMA input selection procedure can be used to determine the inputs.

Finally, we have compared the proposed hybrid methods with pure forecasting techniques, ARIMA and VAR. Model specifications of these two methods have been done in Sections 3.1 and 3.2. The following table (Table 11) shows the MSE of the pure techniques.

Comparison of the hybrid methods with pure techniques reveals that the hybrid method outperforms the pure ARIMA and VAR models in terms of MSE error (see Tables 10 and 11). By using hybrid forecasting

Table 11

MSE of pure forecasting techniques (ARIMA and VAR) for testing set

Exchange rates	ARIMA	VAR
EURO/Dollar	0.001874	0.000039
GBP/Dollar	0.004072	0.000081
JPY/Dollar	6.853610	0.361648
AUD/Dollar	0.001108	0.000071

techniques, we try to avoid the weakness of pure techniques.

4. Conclusions

We try to combine parametric and nonparametric techniques in order to obtain a better performance for exchange rate forecasting. In addition to this, comparison of two nonparametric models, ANN and SVR, is given with two input selection techniques, ARIMA and VAR, respectively. Parametric and nonparametric techniques have some advantages and disadvantages. For example, parametric techniques are based on specific assumptions and these assumptions are not satisfied or partially satisfied in real world problems. Because of this, we use parametric techniques to identify the number of previous values of dependent and independent variables. On the other hand, nonparametric techniques do not have any restricted assumption like parametric techniques. For this reason, they are applied to estimate the parameters of the mathematical models.

Experiments showed that SVR method outperforms the MLP networks for each input selection algorithm. This can be explained by the formulation of the SVR and MLP networks. SVR method uses a quadratic programming problem which is convex and has a global optimum solution. On the other hand, MLP networks use the backpropagation algorithm to minimize the network error. The problem is nonconvex and it is hard to find the global optimum.

Comparison of the input selection process reveals different results. The best selection procedure depends on the training algorithms. If we want to use an MLP network, it is better to use VAR technique to determine the inputs. On the other hand, ARIMA input selection technique gives the best results if SVR method is employed for training. All the comparisons are made by using MSE and MAE error of the testing set.

A different approach has been applied to exchange rate forecasting. This hybrid technique provides very promising results. The next step would be to develop trading strategies. Our goal is to show that combination of parametric and nonparametric techniques is as good as pure techniques.

References

- [1] T. Bollerslev, Generalized autoregressive conditional heteroskedasticity, *Journal of Econometrics* 31 (1986) 307–327.
- [2] C. Brooks, Linear and non-linear (non-)forecastability of high frequency exchange rates, *Journal of Forecasting* 15 (1997) 125–145.
- [3] C.J.C. Burges, A tutorial on support vector machines for pattern classification, *Data Mining and Knowledge Discovery* 2 (2) (1998) 121–167.
- [4] O. Chapelle, V. Vapnik, O. Bousquet, S. Mukherjee, Choosing multiple parameters for support vector machines, *Machine Learning* 46 (13) (2002) 131–159.
- [5] A. Chen, M.T. Leung, H. Daouk, Application of neural networks to an emerging financial market: forecasting and trading the Taiwan stock index, *Computers Operations Research* 30 (2003) 901–923.
- [6] A.-S. Chen, M.T. Leung, Regression neural network for error correction in foreign exchange forecasting and trading, *Computers and Operations Research* 31 (2004) 1049–1068.
- [7] S.-H. Chun, S.H. Kim, Impact of momentum bias on forecasting through knowledge discovery techniques in the foreign exchange market, *Expert Systems with Applications* 24 (2003) 115–122.
- [8] R. Collobert, S. Bengio, Svmtorch: support vector machines for largescale regression problems, *Journal of Machine Learning Research* 1 (2001) 143–160.
- [9] C. Cortes, V. Vapnik, Support vector networks, *Machine Learning* 20 (1995) 273–297.
- [10] N. Cristianini, C. Campbell, J. ShaweTaylor, Dynamically adapting kernels in support vector machines, *NIPS* 1998 (1998) 204–210.
- [11] J.T. Davis, A. Episcopos, S. Wettimuny, Predicting direction shifts on Canadian–US exchange rates with artificial neural networks, *International Journal of Intelligent Systems in Accounting* 10 (2001) 83–96.
- [12] S., Demirbas, Cointegration Analysis—Causality Testing and Wagner's Law: The Case of Turkey, 1950–1990, Annual Meeting of the European Public Choice Society (April 7–10, 1999).
- [13] F.X. Diabold, J. Gardeazabal, K. Yilmaz, On cointegration and exchange rate dynamics, *Journal of Finance* 49 (1994) 727–735.
- [14] Z. Ding, C.W.J. Granger, Modeling volatility persistence of speculative returns: a new approach, *Journal of Econometrics* 73 (1996) 185–215.
- [15] Z. Ding, C.W.J. Granger, R.F. Engle, A long memory property of stock market returns and a new model, *Journal of Empirical Finance* 1 (1993) 83–106.
- [16] R.F. Engle, Autoregressive conditional heteroscedasticity with estimates of the variance of the United Kingdom inflation, *Econometrica* 50 (1982) 987–1008.
- [17] R.F. Engle, C.W. Granger, Co-integration and error correction: representation, estimation and testing, *Econometrica* 55 (1987) 251–276.
- [18] M. Fernandes, Non-linearity and exchange rates, *Journal of Forecasting* 17 (1998) 497–514.
- [19] J. Galindo, A framework for comparative analysis of statistical and machine learning methods: an application to the black scholes option pricing equations, Technical Report, Banco de Mexico, Mexico, DF, 1998 (04930).
- [20] J.D. Hamilton, *Time Series Analysis*, Princeton Univ. Press, 1994.
- [21] S. Haykin, *Neural Networks: A Comprehensive Foundation*, MacMillan Publishing Company, New York, 1994.
- [22] J.M. Hutchinson, A.W. Lo, T. Poggio, A nonparametric approach to pricing and hedging derivative securities via learning networks, *The Journal of Finance* XLIX (3) (1994) 851–889.
- [23] H. Ince, B. Trafalis, Short term forecasting with support vector machines and application to stock price prediction, in: Dagli, Buczak, Ghosh, Embrechts, Ersoy (Eds.), *Smart Engineering*

- System Design: Neural Networks, Fuzzy Logic, Evolutionary Programming, Data Mining and Complex Systems, ASME Press, 2003, pp. 737–746.
- [24] S. Johansen, Estimation and hypothesis of cointegration vectors in Gaussian vector autoregressive models, *Econometrica* 59 (1991) 1551–1580.
- [25] L. Kilian, M.P. Taylor, Why is it so difficult to beat random walk forecast of exchange rates? *Journal of International Economics* 60 (2003) 85–107.
- [26] V. Kodogiannis, A. Lolis, Forecasting financial time series using neural network and fuzzy system-based techniques, *Neural Computing and Applications* 11 (2002) 90–102.
- [27] Y.J. Lee, O.L. Mangasarian, Rsvm: reduced support vector machines, CD Proceedings of the First SIAM International Conference on Data Mining, 2001.
- [28] X. Li, C.-L. Ang, R. Gray, An intelligent business forecaster for strategic business planning, *Journal of Forecasting* 18 (1999) 181–204.
- [29] F. Lisi, R.A. Schiavo, A comparison between neural networks and chaotic models for exchange rate prediction, *Computational Statistics and Data Analysis* 30 (1999) 87–102.
- [30] A.K. Nag, A. Mitra, Forecasting daily foreign exchange rates using genetically optimized neural networks, *Journal of Forecasting* 21 (2002) 501–511.
- [31] E. Osuna, R. Freund, F. Girosi, Training support vector machines: an application to face detection, *Proc. Computer Vision and Pattern Recognition '97*, 1997, pp. 130–136.
- [32] J. Platt, Fast training of support vector machines using sequential minimal optimization, in: B. Schölkopf, C.J.C. Burges, A.J. Smola (Eds.), *Advances in Kernel Methods: Support Vector Learning*, MIT Press, 1999, pp. 185–208.
- [33] M. Pontil, A. Verri, Properties of support vector machines, Technical Report, Massachusetts Institute of Technology, Artificial Intelligence Laboratory, 1997.
- [34] B. Schölkopf, A.J. Smola, *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*, The MIT Press, Cambridge, Massachusetts, 2002.
- [35] T.B. Trafalis, Artificial neural networks applied to financial forecasting, in: C.J.C. Dagli, Buczak, Ghosh, Embrechts, Ersoy (Eds.), *Smart Engineering System Design: Neural Networks, Fuzzy Logic, Evolutionary Programming, data Mining and Complex Systems*, ASME Press, 1999, pp. 1049–1054.
- [36] T.B. Trafalis, H. Ince, Support vector machine for regression and applications to financial forecasting, *Neural Networks*, 2000. IJCNN 2000, Proceedings of the IEEEINNSSENNNS International Joint Conference, vol. 6, IEEE, 2000, pp. 348–353.
- [37] T.B. Trafalis, H. Ince, T. Mishina, Support vector regression in option pricing, *Proceedings of Conference on Computational Intelligence and Financial Engineering (CIFer 2003)*, 2003 (March 20–23) Hong Kong, 2003.
- [38] A. Trapletti, A. Geyer, F. Leisch, Forecasting exchange rates using cointegration models and intra-day data, *Journal of Forecasting* 21 (2002) 151–166.
- [39] R. Tsaih, Sensitivity Analysis, *Neural Networks and the Finance*, 1999, pp. 3830–3835.
- [40] R.J. Vanderbei, Interior point methods: algorithms and formulations, *ORSA Journal on Computing* 6 (1) (1995) 32–34.
- [41] R.J. Vanderbei, LOQO an interior point code for quadratic programming, Technical Report, Statistics and Operations Research, Princeton University, 1998 (SOQ-94-15).
- [42] V. Vapnik, *The Nature of Statistical Learning Theory*, Springer Verlag, 1995.
- [43] J. Vilasuso, Forecasting exchange rate volatility, *Economic Letters* 76 (2002) 59–64.
- [44] Z. Vojinovic, V. Kecman, R. Seidel, A data mining approach to financial time series modeling and forecasting, *International Journal of Intelligent Systems, Finance & Management* 10 (2001) 225–239.
- [45] J. Yao, C.H. Tan, A case study on using neural networks to perform technical forecasting of forex, *Neurocomputing* 34 (2000) 79–98.

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